

Jahangirnagar University



Assignment Topic

Developing a Student Information System (SIS) Using Product Breakdown Structure (PBS)

Course Title: Software Project Management and Quality Assurance
Course Code: PMSCS 660

SUBMITTED TO

Dr. Mohammad Zahidur Rahman
Professor, Department of Computer Science and Engineering

SUBMITTED BY

Md. Habibur Rahman
Id No: CSE202601013
Department of Computer Science and Engineering

Date: 11th June 2026

Developing a Student Information system (SIS) Using product Breakdown Structure (PBS):

1. Objectives:

The main objectives of developing the Student Information System (SIS) are:

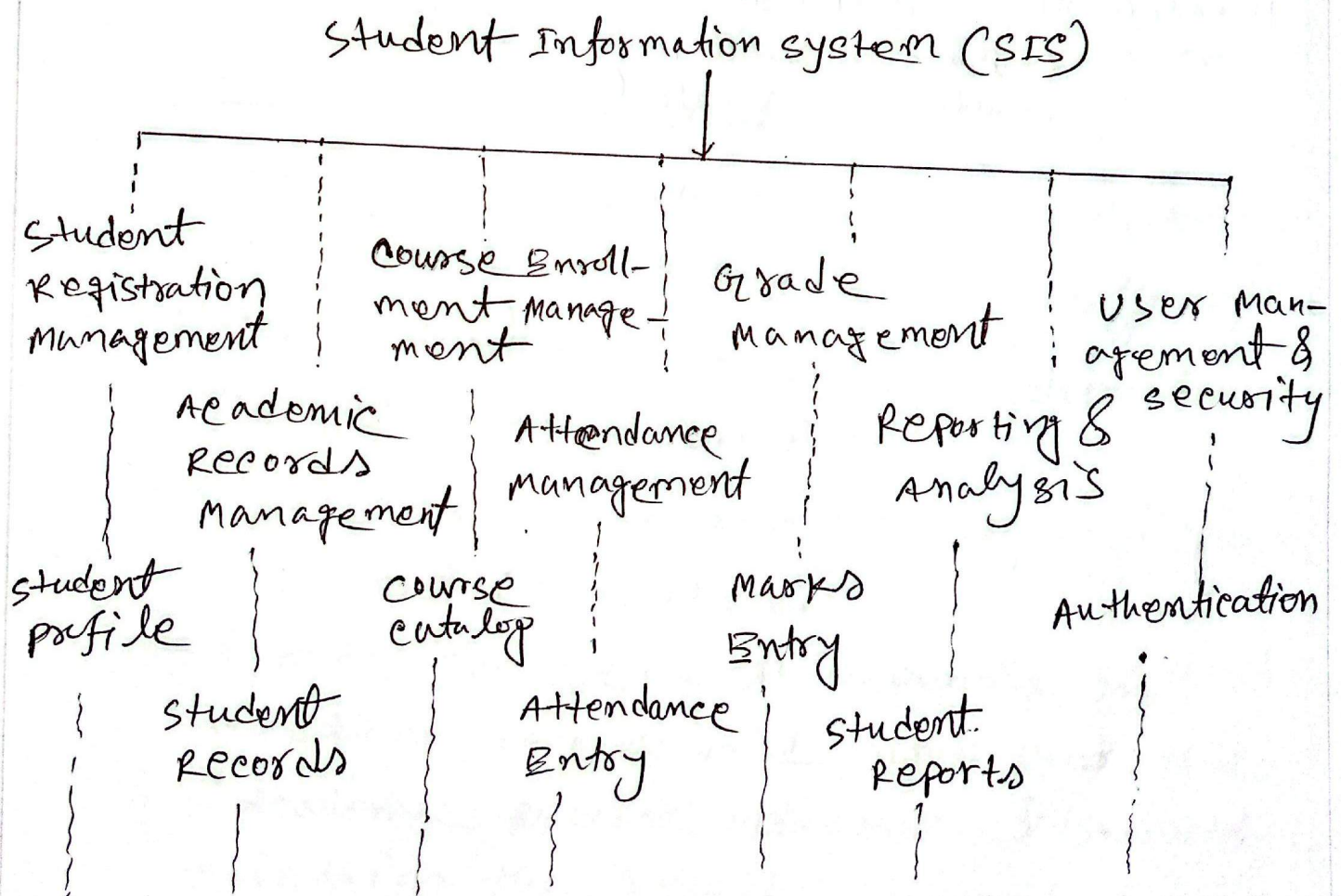
- a) To centralize information management system (SIS) in a single system.
- b) To automate academic and administrative process.
- c) To improve the accuracy and security of student data.
- d) To facilitate efficient course enrollment attendance tracking and grade management.
- e) To provide timely and accurate reports for students, faculty and administrative.
- f) To enhance communication and information accessibility within the institutions.

2. Major components of the Student Information System:

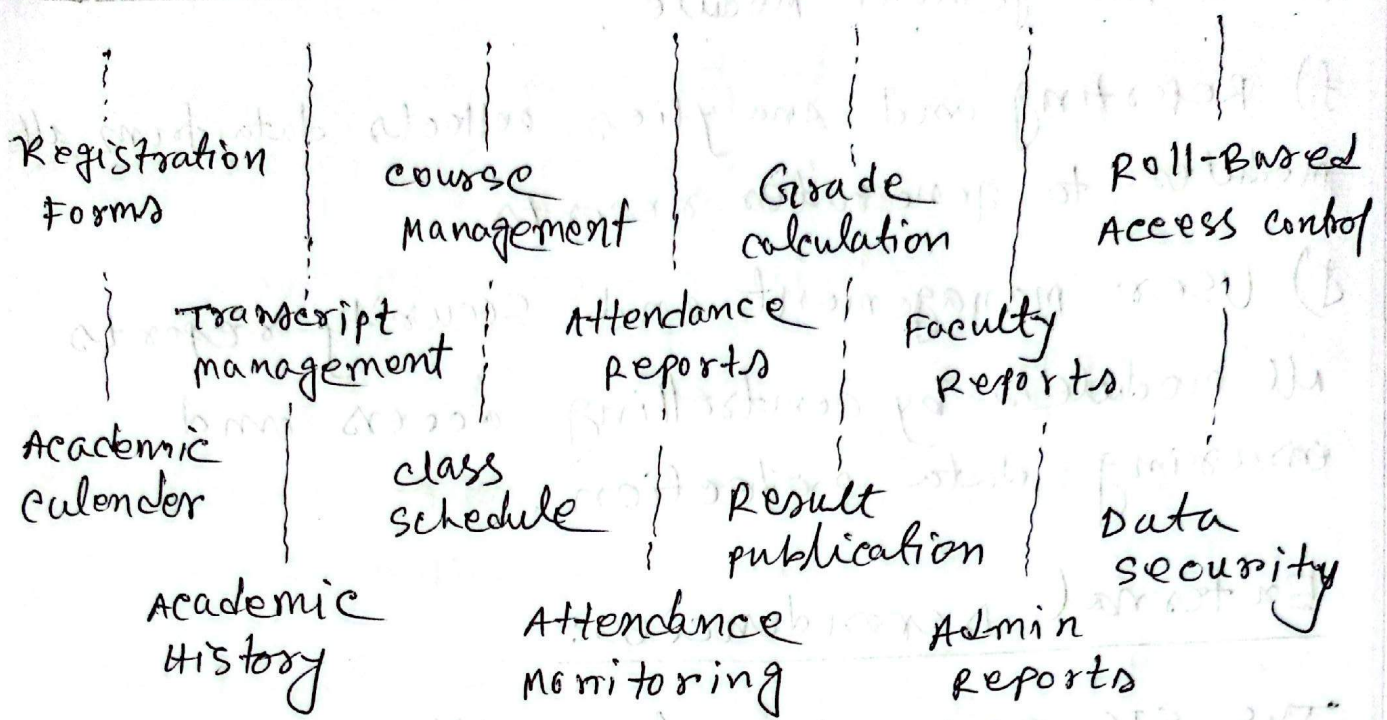
The proposed Student Information system (SIS) consists of the following major components:

- a) Student Registration management.
- b) Academic Records management.
- c) Course enrollment management.
- d) Attendance management.
- e) Grade management.
- f) Reporting and Analysis.
- g) User management and security.

3. Product Breakdown Structure (PBS):



(P.T. 01)



4. Dependency Analysis:

The components of the student information system are interconnected

- Student registration management must be completed before course Enrollment.
- Course Enrollment management depends on student registration data.
- Attendance management requires course enrollment information.
- Grade management depends on attendance records and enroll courses.
- Academic records management stores and maintains the results generated by the

Grade management module .

f) Reporting and Analytics collects data from all modules to generate reports .

g) User management and security reports all modules by controlling access and ensuring data protection .

External dependencies:

The SIS may integrate with :

a) Learning management system (LMS)

b) Library management system (LMS)

c) Payment gateway .

d) Email and SMS services .

e) Institutional Student Information System (ISIS)

5. proposed integration points :

Learning management system :

students can be access library services using the same institutional account .

Payment Gateway:

Supports online payment of tuition fees and other academic charges.

Email and SMS services:

Provide notifications regarding registration, attendance, examination results and important announcements.

Institutional Student Information System (SIS)

Allows secure data exchange between the departmental SIS and the institution-wide information system.

6. Additional Feature:

The following features may further enhance the system:

1. Mobile Application supports.
2. Online Advising system.
3. Student performance dashboard.
4. Automated Notifications
5. Real-Time Analytics.
6. Role-Based Access control.

7. Conclusion:

The proposed Student Information System (SIS) provides a centralized platform for managing student information, academic records, attendance, grades and courses enrollment.

The product Breakdown Structure (PBS) helps identify all major system components and their sub-components, ensuring a systematic development process. The system improves efficiency, accuracy and accessibility while supporting future integration with other institutional systems.